Increasing Developer Productivity

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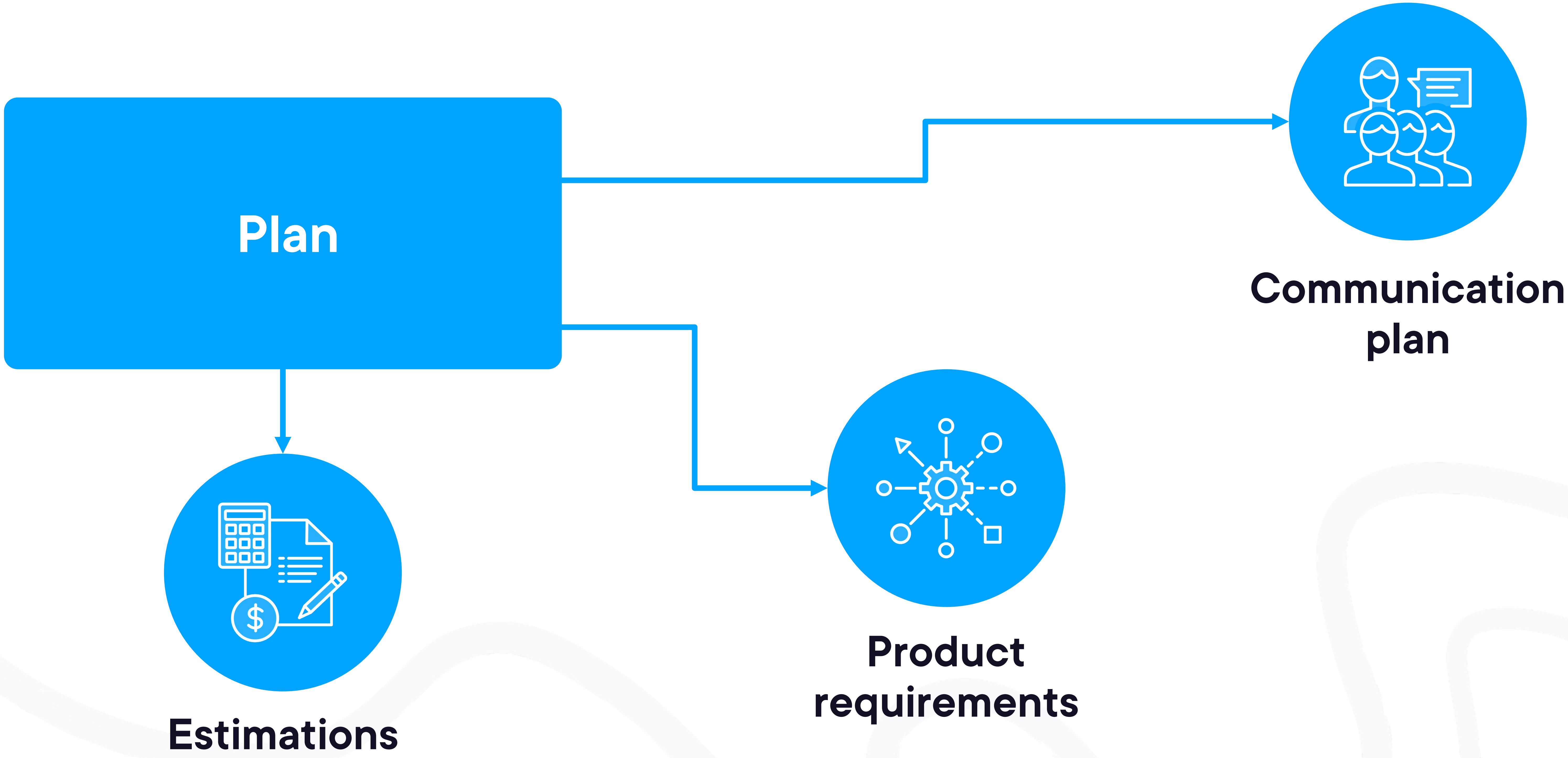
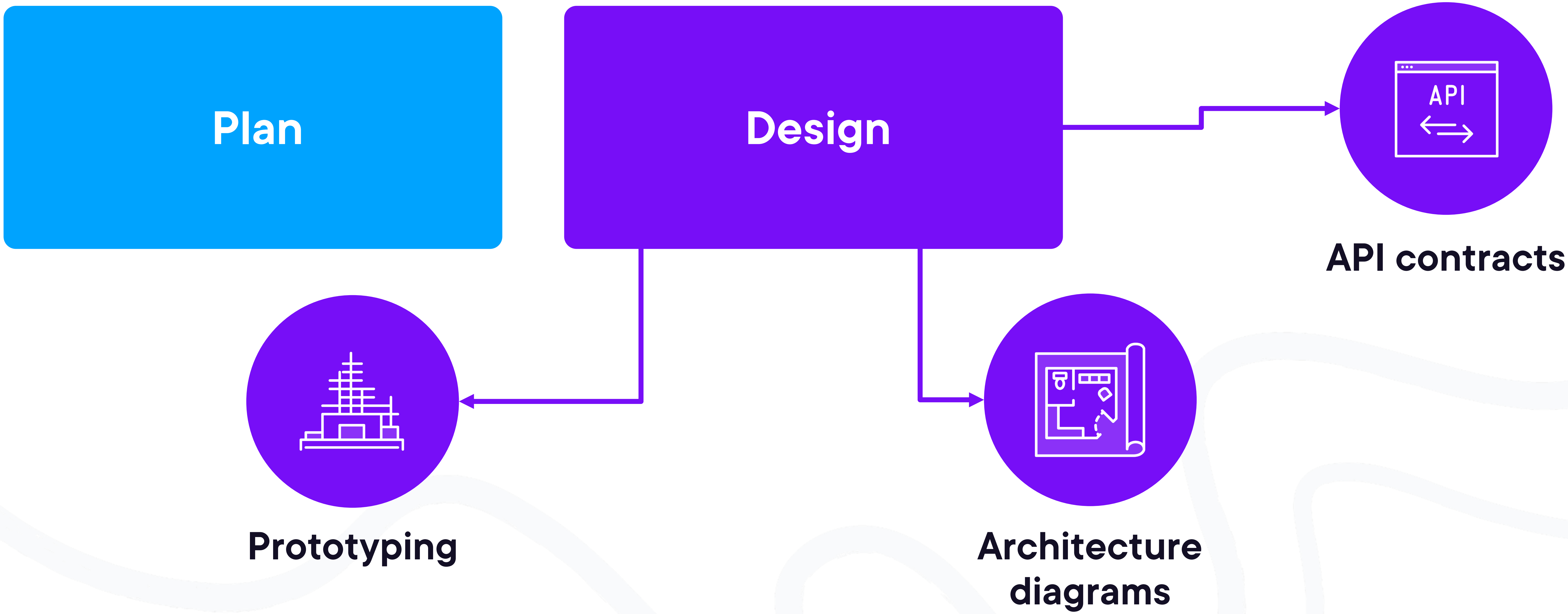
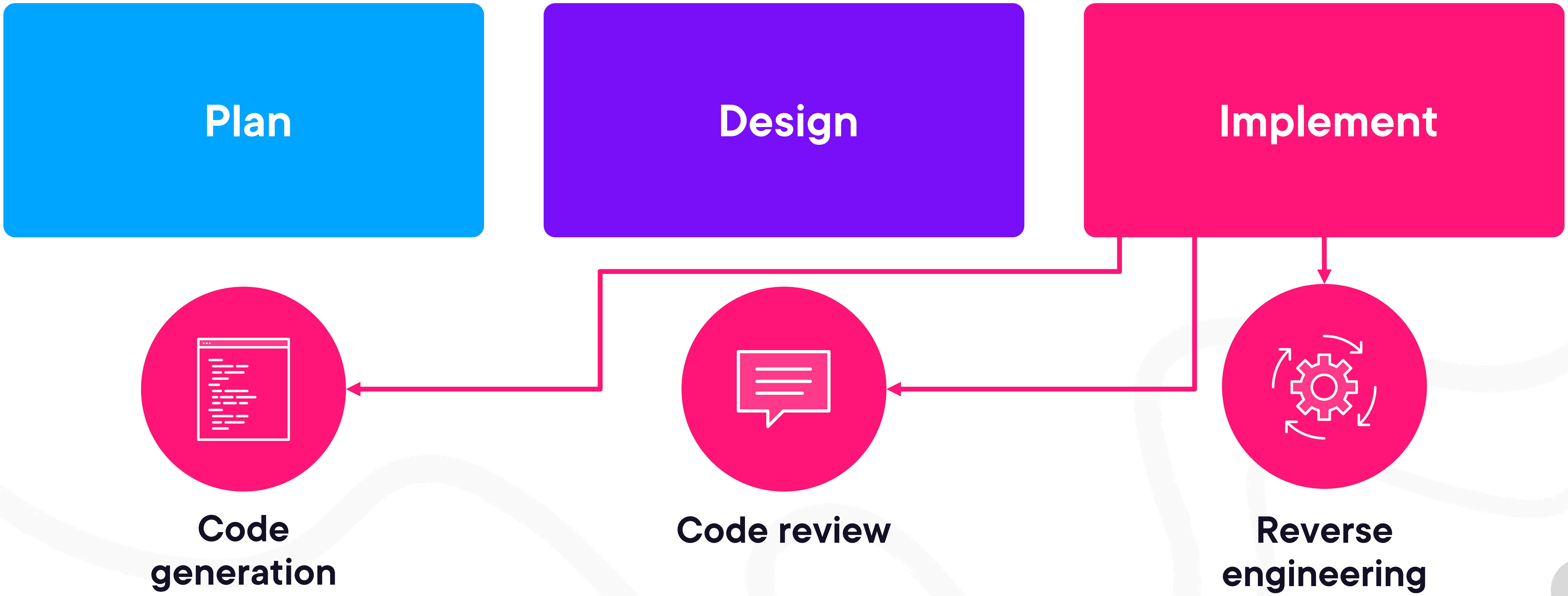
Software Development Lifecycle



Software Development Lifecycle



Software Development Lifecycle



Software Development Lifecycle

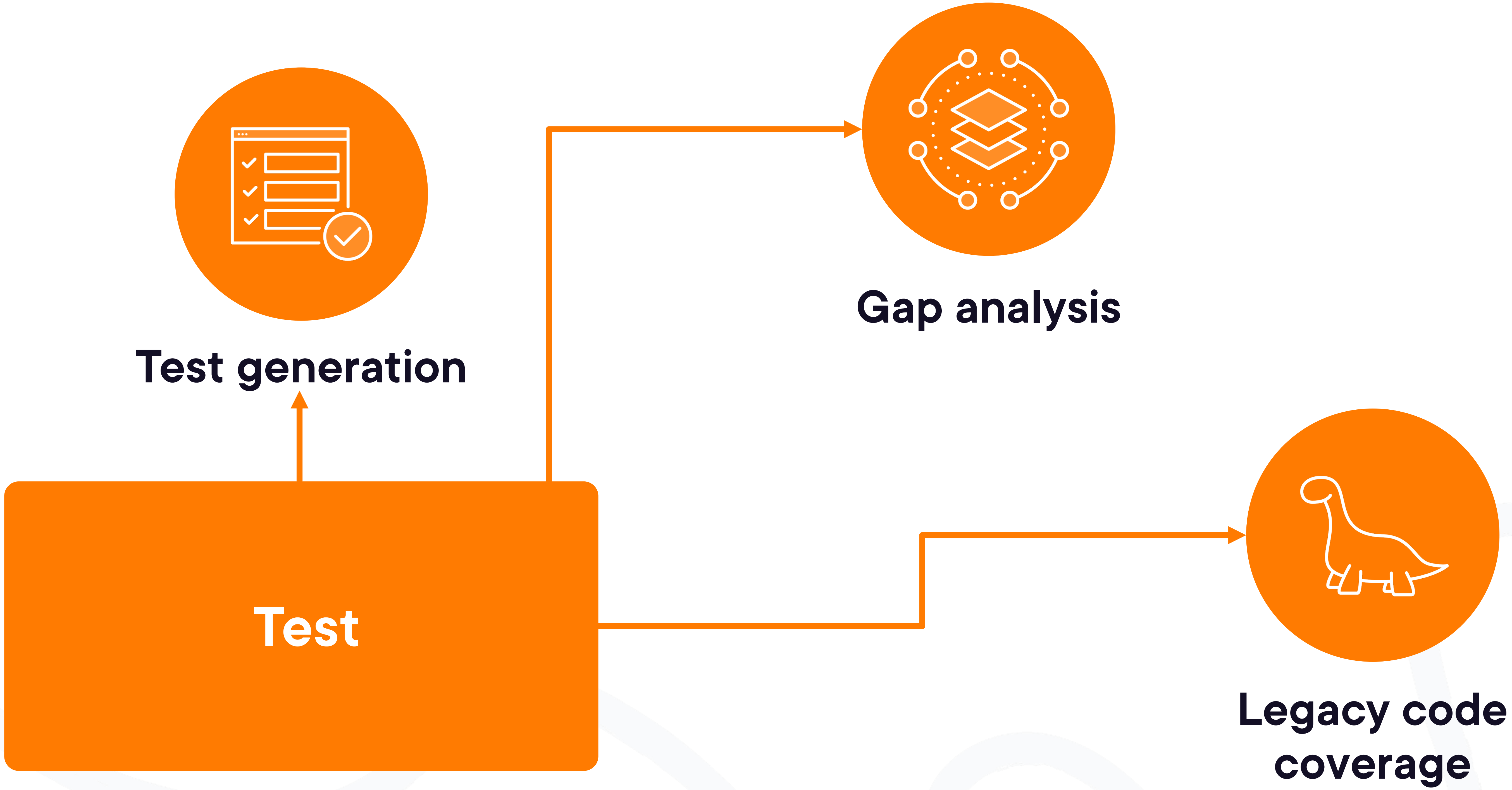
Plan

Design

Implement

Test

Software Development Lifecycle



Software Development Lifecycle

Plan

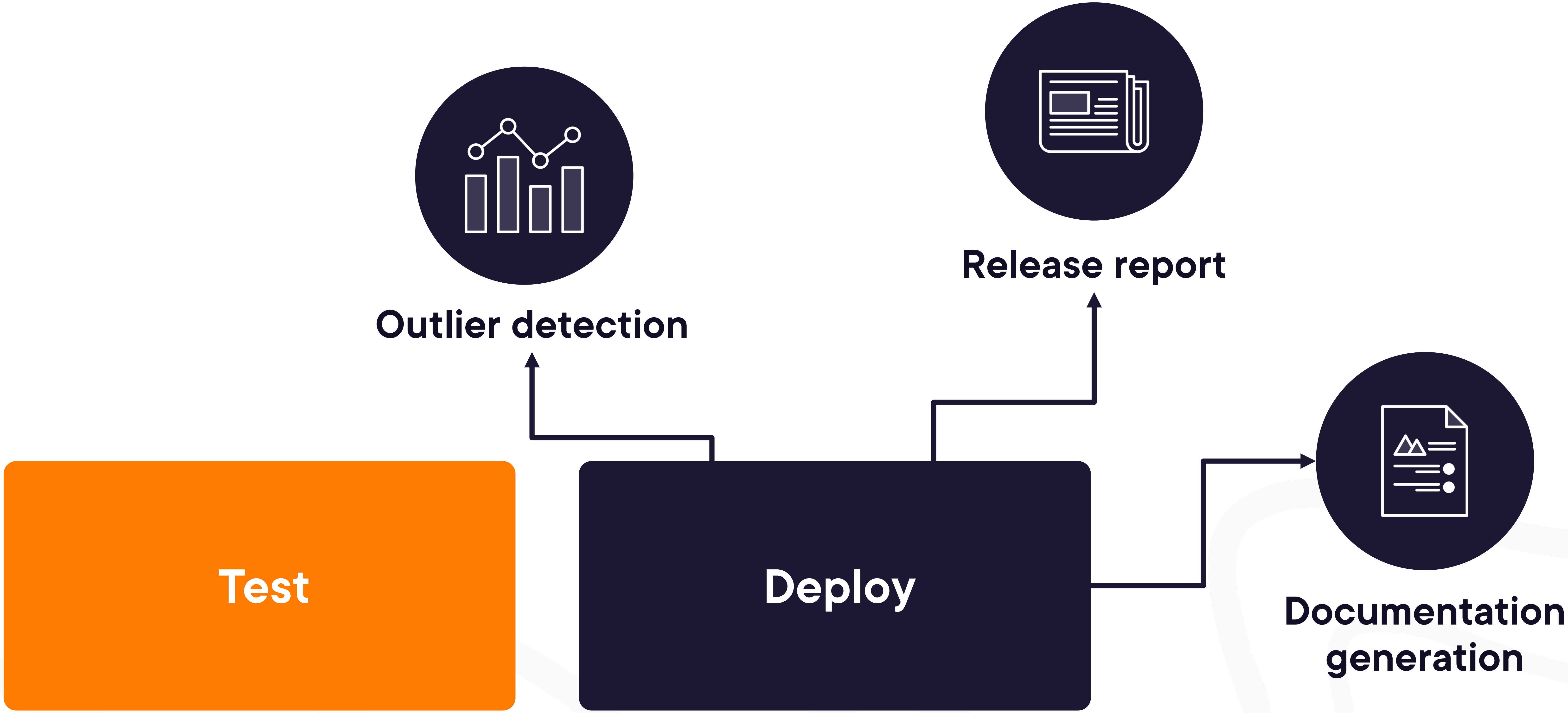
Design

Implement

Test

Deploy

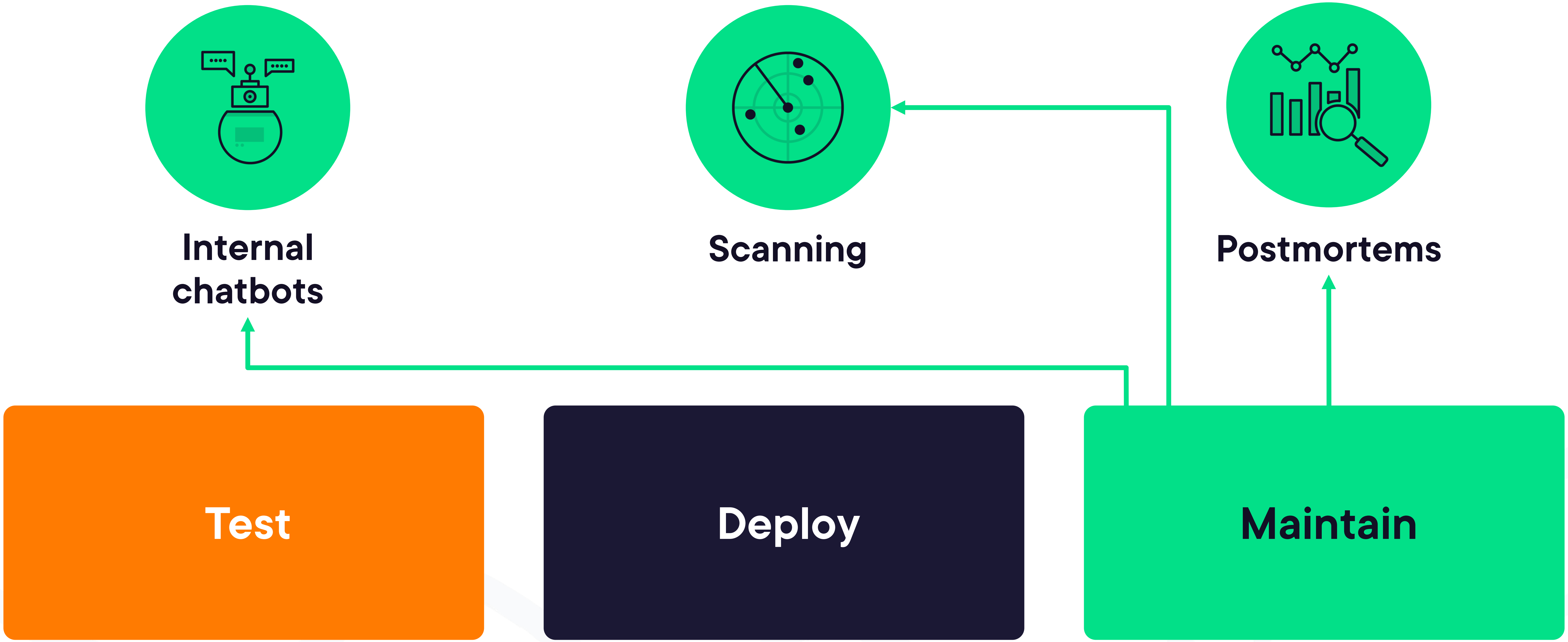
Software Development Lifecycle



Software Development Lifecycle

Plan	Design	Implement
Test	Deploy	Maintain

Software Development Lifecycle



Agentic Coding

Code Generation

LLMs

AI coding assistants

Agentic coding

AI Coding Assistant

```
function calculateTotal(items) {
        let total = 0;
```

Autocomplete

Chat

AI Coding Assistant

```
function calculateTotal(items) {
        let total = 0;
        items.forEach(item => total += item.price * item.quantity);
        return total;
}
```

Autocomplete

Chat

AI Coding Assistant

```
function calculateTotal(items) {
        let total = 0;
        items.forEach(item => total += item.price * item.quantity);
        return total;
}
```

Autocomplete

Build a JavaScript function that calculates the total price of the items

Chat

AI Coding Assistant

```
function calculateTotal(items) {
        let total = 0;
        items.forEach(item => total += item.price * item.quantity);
        return total;
}
```

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Autocomplete

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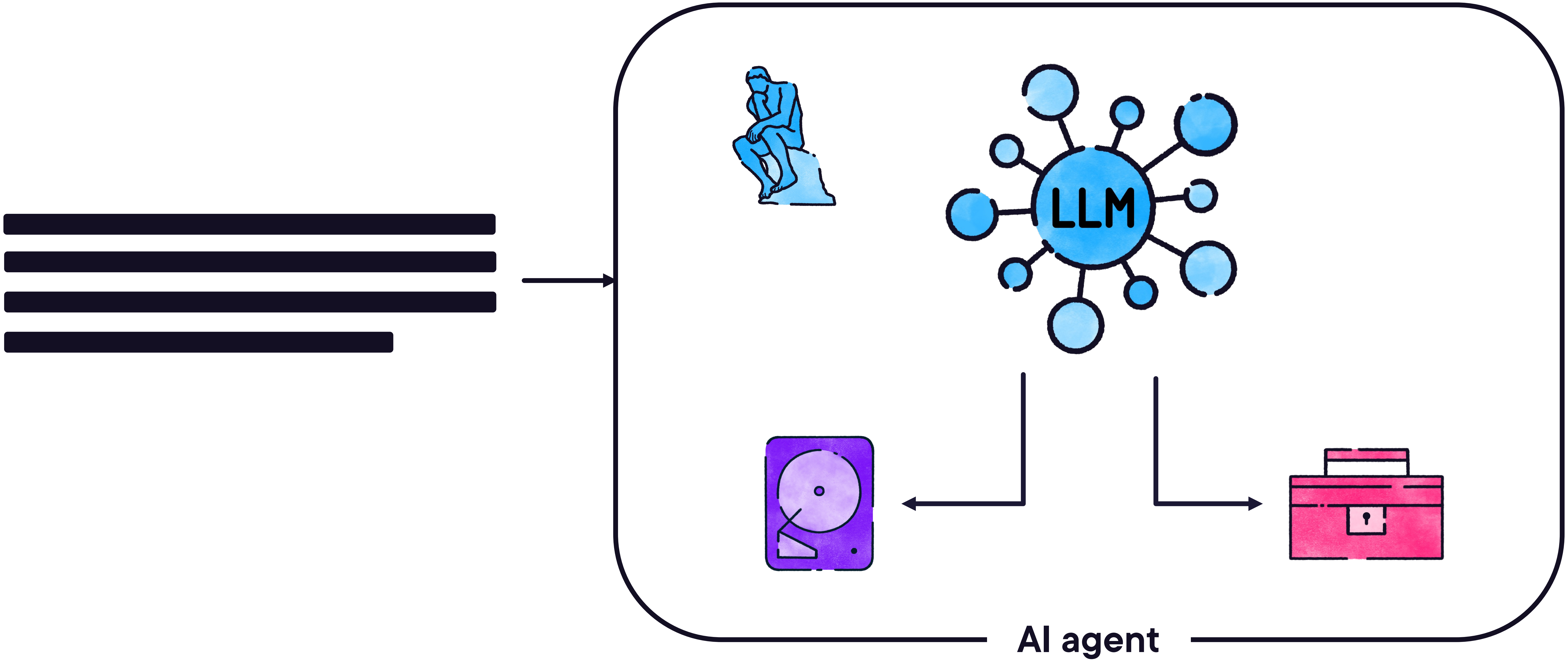
Agentic Coding



Modality

IDE Plugin

IDE Plugins

GitHub Copilot

Tabnine

Amazon Q

JetBrains AI

Modality

IDE Plugin

IDE

IDE

Cursor

Windsurf

Modality

IDE Plugin

IDE

CLI

CLI

Claude Code

Aider

OpenAI Codex CLI

Gemini CLI

Modality

IDE Plugin

IDE

CLI

Web interface

Web Interface

OpenAI Codex

Jules by Google

GitHub Copilot Workspace

Use Cases

Code generation

Test generation

Reverse engineering and debugging

Documentation

Refactoring

Global Rules

Global Rules

Security and compliance

Coding standards

Safeguards

Configuration

Memory

Tools

Permissions

MCP Servers

Global Rules

Enterprise policy (shared)

Global (just you)

Project (shared)

Project (just you)

Demo: Global Rules

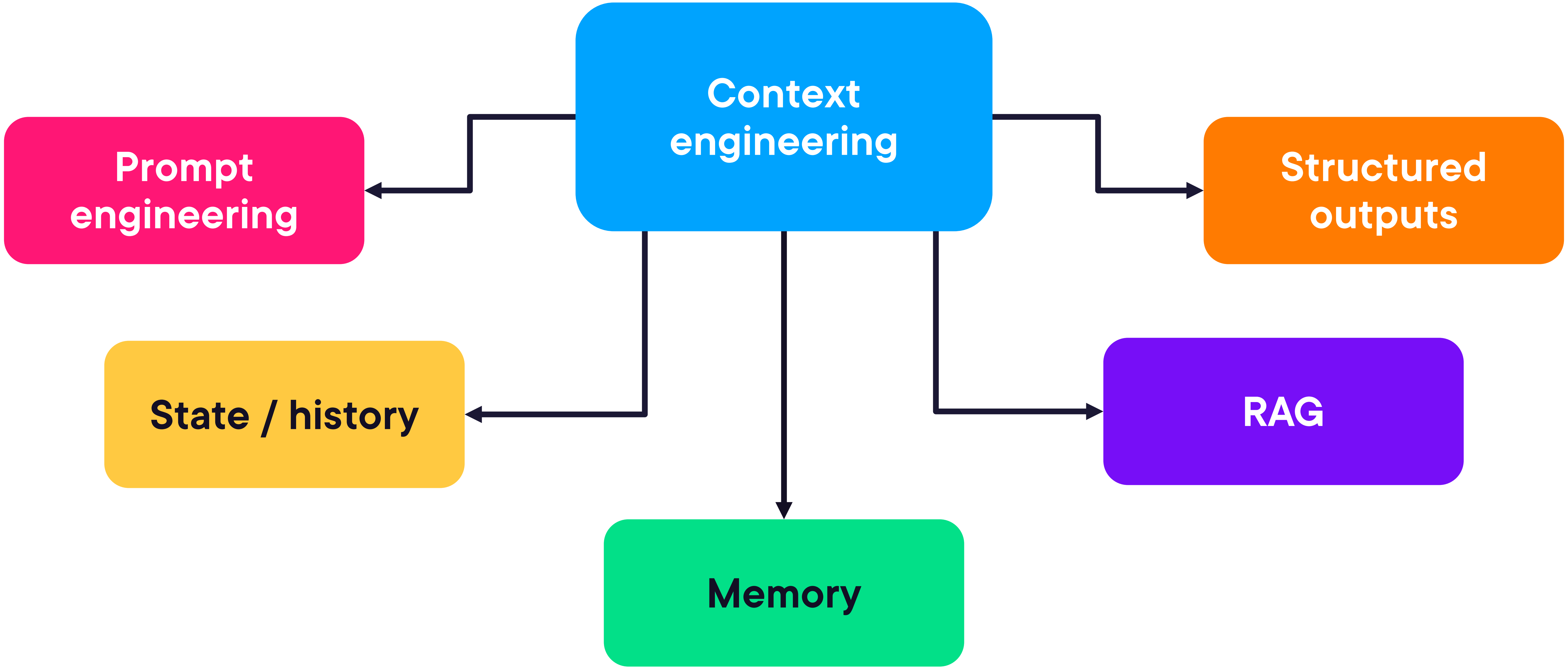
Demo: One-shot Prompt

Vibe Coding

An approach to producing software by using artificial intelligence (AI), where a person describes a problem in a few natural language sentences as a prompt to a large language model (LLM) tuned for coding.

Citation: wikipedia.org/wiki/Vibe_coding

Context Engineering



Demo: Reverse Engineering

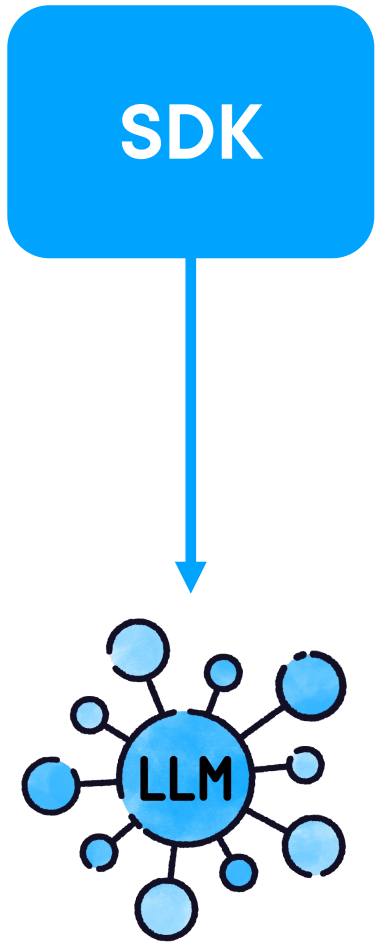
Demo: Test Creation

Demo: Documentation

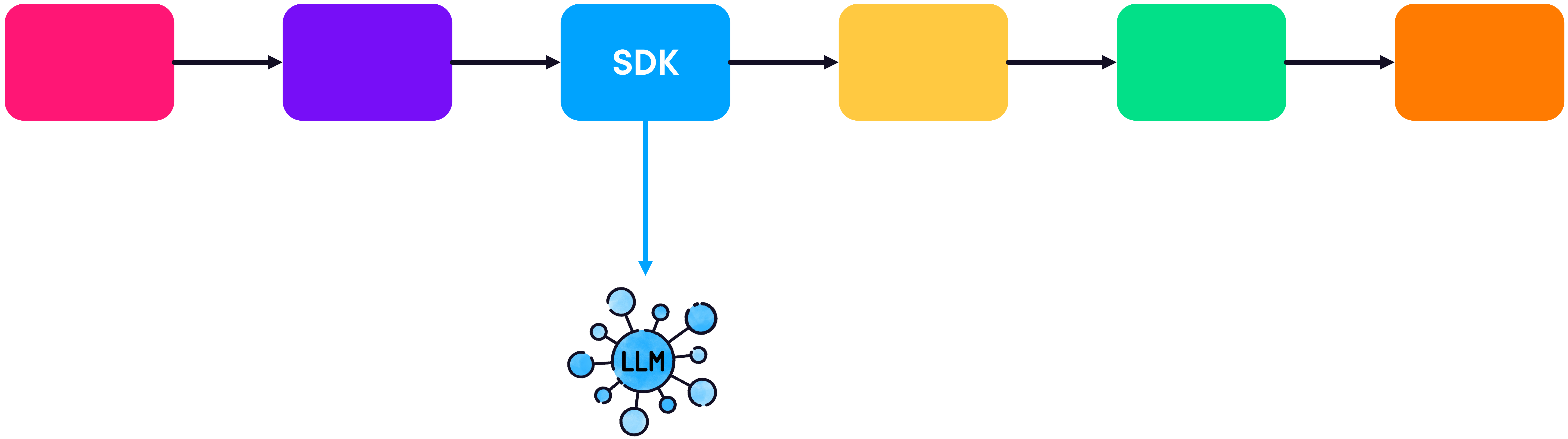
Demo: Refactoring

CI / CD Pipelines

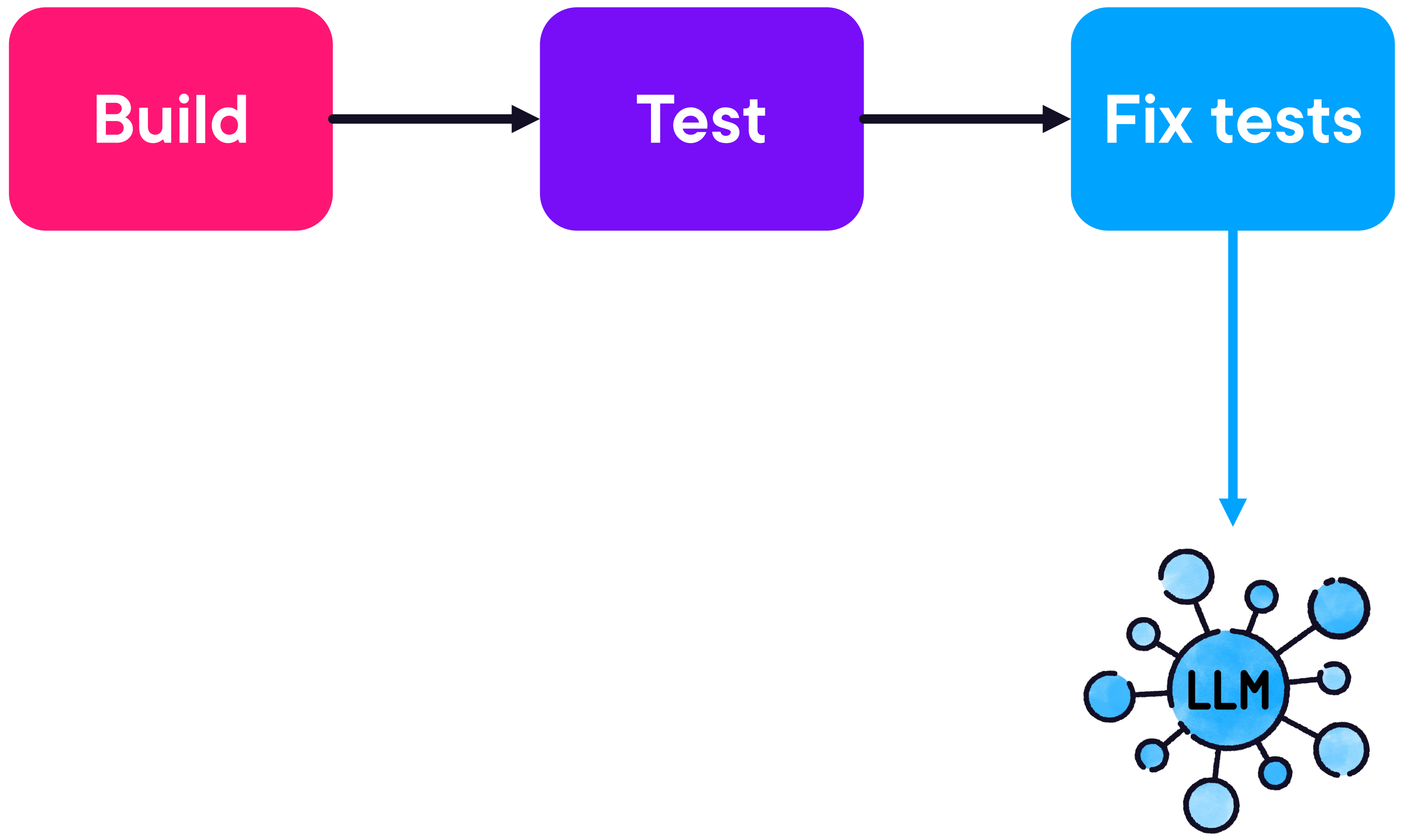
LLM SDK



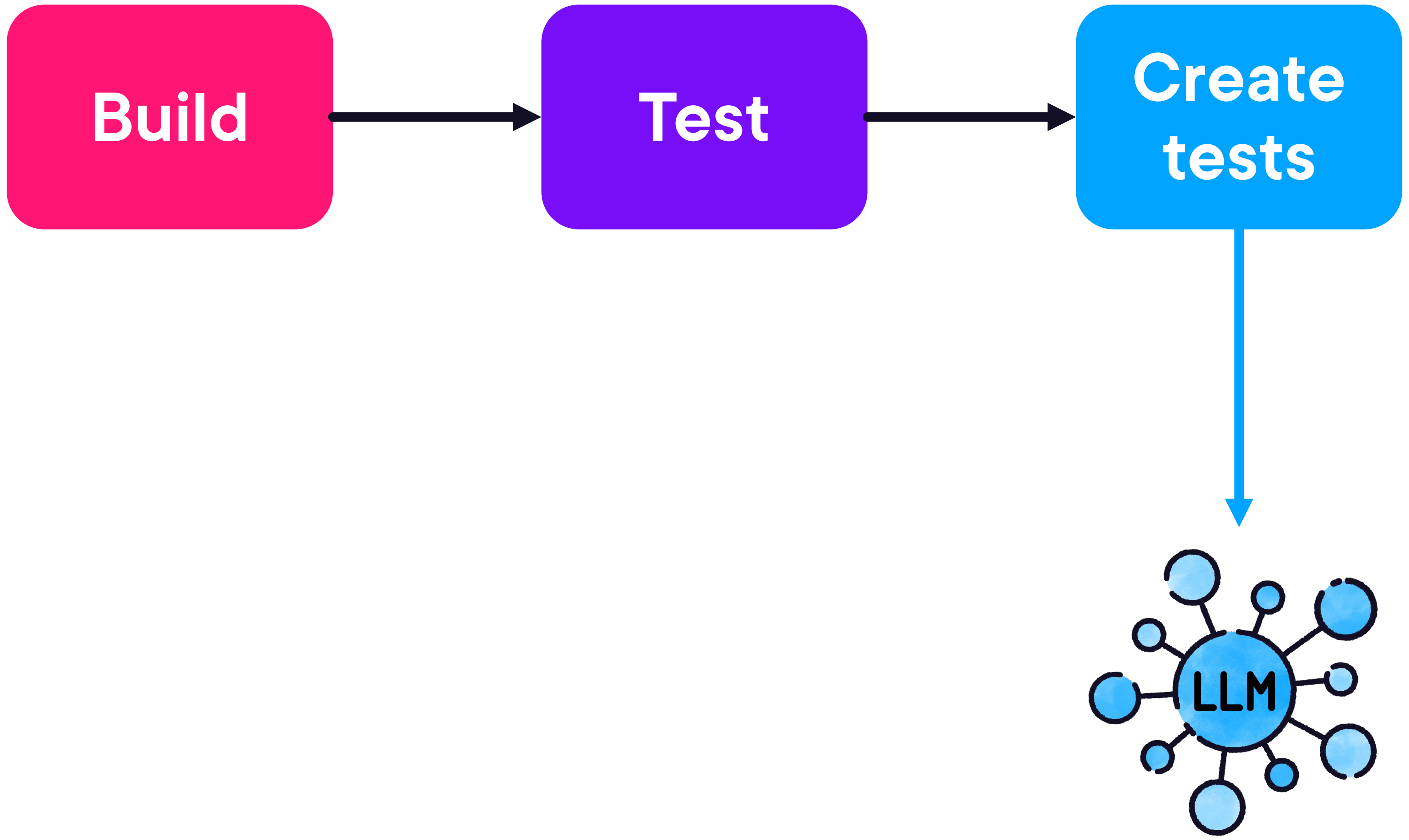
CI Pipeline



CI Pipeline



CI Pipeline



Use Cases

Test generation

Code implementation

Code review

Report generation

Documentation